

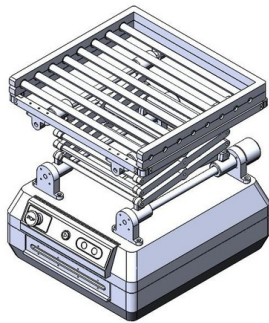
CAD DRAFTING / MECHANICAL DESIGN / FABRICATION

THAM WENG HUNG

Drafter & Product Design Engineer

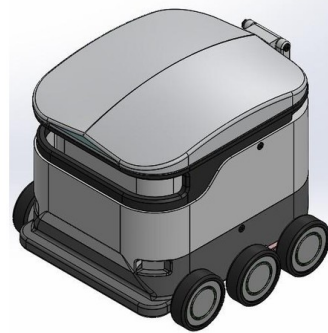
Fabrication-ready documentation. Practical product design. Clear handover.

ASSEMBLY STUDY / 01



SOLIDWORKS / 3D ASSEMBLY / TECHNICAL DOCUMENTATION

PRODUCT STUDY / 02



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01 / PROFILE

Design that moves from screen to shop floor.

Mechanical and project engineer with more than four years of experience across CAD drafting, product development, technical sales, fabrication support and infrastructure commissioning. I prepare clear, controlled documentation for mechanical parts, assemblies and custom metalwork, supported by practical understanding of cutting, bending, rolling, machining, assembly and site handover.

CONFIDENTIALITY Professional work from Petracarbon and Metalex is presented through responsibilities, deliverables and engineering scope only. Employer-owned drawings and proprietary geometry are not reproduced. Visual case studies in this portfolio are selected academic work that can be shared.

CORE SERVICES

2D CAD & DETAILING	Fabrication drawings, shop drawings, sections, details, revisions and as-built updates.
3D PARTS & ASSEMBLIES	SolidWorks component and assembly modelling, packaging studies and exploded views.
SHEET METAL	Panels, frames, cut profiles, bend information and fabrication drawing packages.
CAD-TO-CNC SUPPORT	True-scale geometry and DXF/DWG preparation for laser or plasma cutting.
TECHNICAL HANDOVER	Organised native files, neutral formats, PDFs, BOMs and concise issue information.

SOFTWARE & OUTPUT

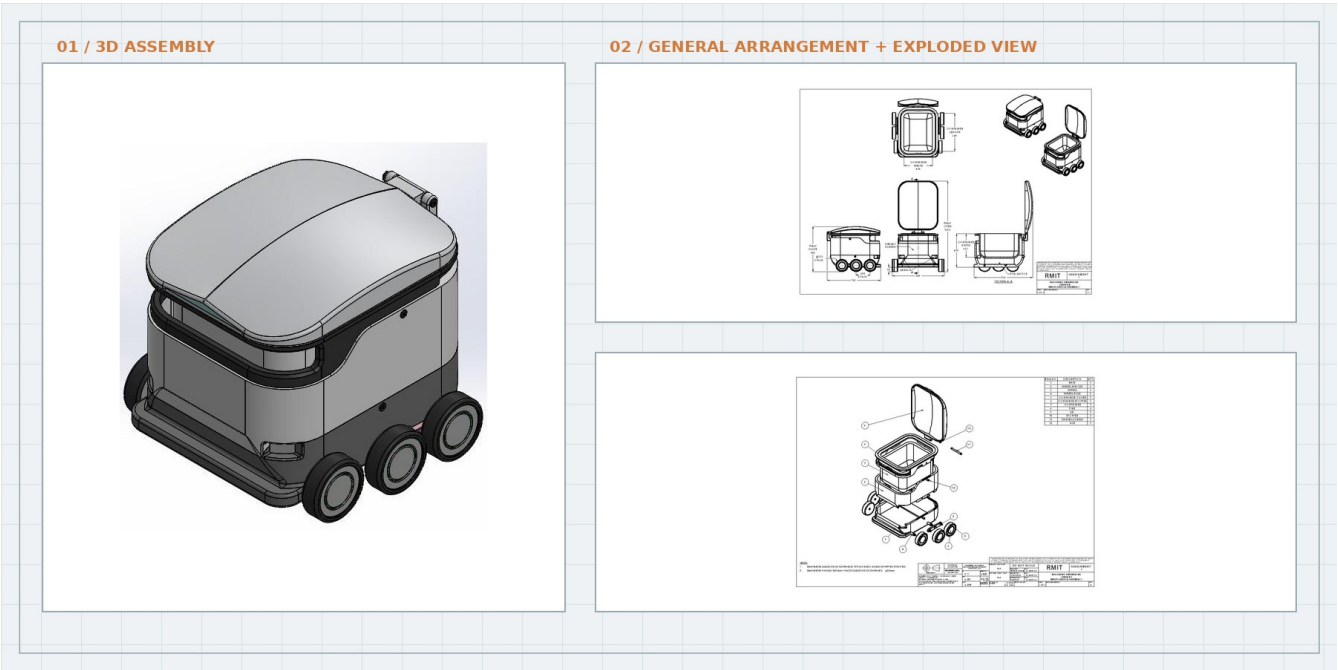
AutoCAD / SolidWorks / Revit / SketchUp / CNC programming & operation

Typical delivery: DWG / DXF / STEP / STL / PDF

02 / ACADEMIC CASE STUDY

Enclosed Mobile Service Platform

Assembly-level product development demonstrating component integration, packaging and technical drawing communication.



Selected academic work / SolidWorks / dimensions shown in millimetres where indicated

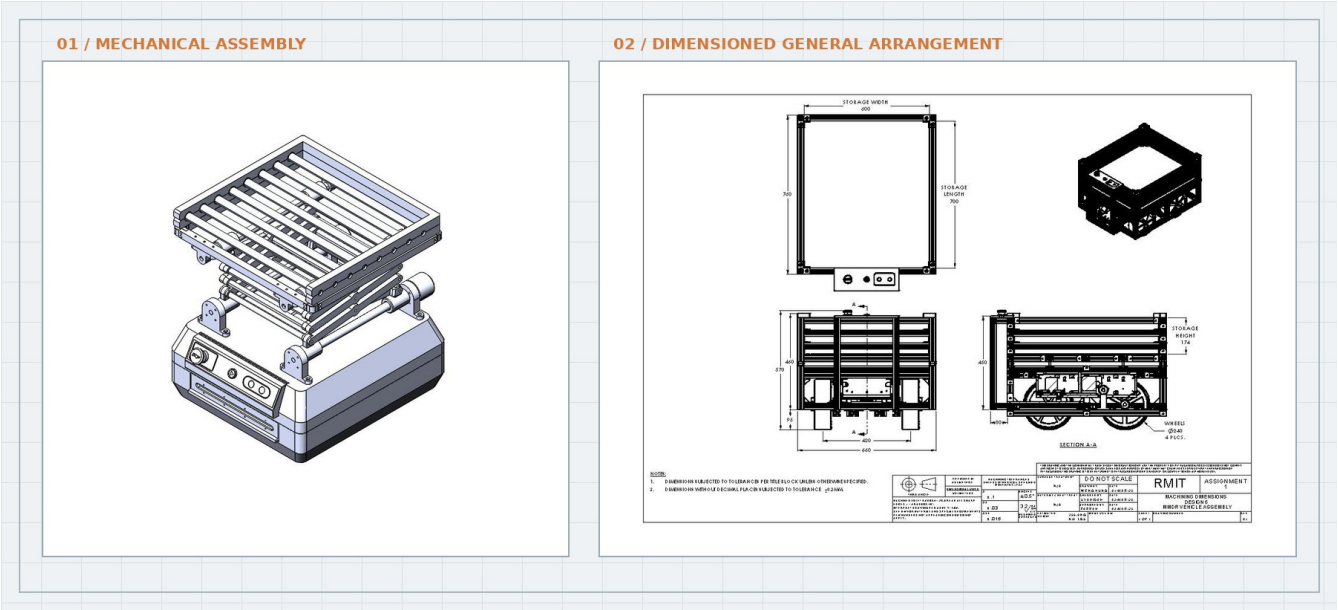
A compact six-wheel platform developed from exterior product form through general arrangement and exploded assembly documentation. The enclosure, hinged lid, container interface, wheel layout and lower chassis were coordinated as one product rather than isolated parts.

ROLE	Mechanical design, 3D assembly modelling and drawing preparation
DESIGN FOCUS	Enclosure geometry / hinged interfaces / wheel arrangement / component packaging
OUTPUT	3D assembly / orthographic and section views / exploded view / item balloons / BOM

03 / ACADEMIC CASE STUDY

Roller-Lift Transfer Platform

A multi-component mechanism combining a roller deck, lift linkage, drive elements and compact mobile base.



Selected academic work / assembly model and general arrangement drawing

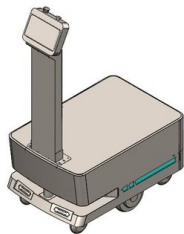
The assembly explores vertical motion, load-transfer geometry and mechanism packaging within a constrained base. The drawing communicates the storage envelope, operating height, wheel layout, internal component relationships and sectional information required to understand the design.

ROLE	Mechanical design, assembly modelling and drawing preparation
DESIGN FOCUS	Scissor linkage / roller deck / structural frame / actuation packaging
OUTPUT	Multi-part assembly / dimensioned GA / section view / stored and operating envelope

04 / MODEL STUDY

Mobile Equipment Platform

An additional SolidWorks study showing enclosure development, compact chassis proportions and equipment mounting.



MOBILE EQUIPMENT PLATFORM

CHASSIS / ENCLOSURE / EQUIPMENT MOUNT

- 01 Compact wheel-module arrangement
- 02 Vertical equipment-mounting structure
- 03 Enclosure and chassis proportion study
- 04 Assembly-level product presentation

Academic product-form and component-packaging study

DRAWING & ASSEMBLY EVIDENCE

3D ASSEMBLY

Coordinated components, interfaces, enclosure and motion-related parts.

GENERAL ARRANGEMENT

Orthographic and isometric views communicating overall form and envelope.

SECTION & DIMENSIONS

Internal relationships, working height and key design dimensions.

EXPLODED VIEW & BOM

Assembly sequence, item balloons, quantities and component identification.

05 / PROFESSIONAL EXPERIENCE

Professional scope, presented without confidential drawings.

The following work demonstrates commercial design exposure and manufacturing responsibility while respecting confidentiality and intellectual-property obligations.

2023-2025 PETRACARBON PTE. LTD.**Product Engineer & Sales Support**

Developed industrial tools and assemblies from initial requirements through CAD definition, engineering documentation and production support.

- Air motors - component design, assembly structure, housing and rotor interfaces.
- Torque-wrench and tensioning systems - mechanical interfaces, adjustment features and assembly documentation.
- Flange-facing machines - from-scratch machine concepts, assemblies and manufacturing drawings.
- Supported CNC requirements, technical revisions and production feedback.

2021-2023 METALEX ENGINEERING PTE. LTD.**Drafter cum Machine Operator**

Translated customer requirements into practical sheet-metal and fabrication drawings, then supported the CAD-to-machine workflow on the shop floor.

- Custom house-number plates and true-scale cut profiles for laser or plasma cutting.
- Door panels, grilles, metal frames and other fabricated products.
- Scaffolding and structural fabrication designs requiring clear member and interface information.
- CNC preparation, bending, rolling and production revisions.

PORTFOLIO POLICY No company drawing, client dimension, proprietary tolerance or production file is included. Additional professional visuals will be added only when written permission or a fully independent, non-proprietary recreation is available.

CURRENT EXPERIENCE**HollySys (Asia Pacific) Pte. Ltd. / Project Engineer / Nov 2025-Present**

Infrastructure installation, testing and commissioning, as-built documentation, reports and site coordination.

06 / DELIVERY

From brief to a usable drawing package.

A practical workflow for one-off drafting tasks, drawing clean-up or larger mechanical design packages.

01	DEFINE	Review references, dimensions, material, manufacturing process, standard, format and deadline.
02	MODEL	Build or clean the 2D/3D geometry and confirm interfaces and assumptions.
03	DETAIL	Prepare views, dimensions, notes, tolerances, sections, callouts and revision information.
04	CHECK	Review geometry, completeness, file naming, print scale and export integrity.
05	HAND OVER	Issue organised native files, neutral formats, PDFs, cut files and supporting notes.

FREELANCE DELIVERABLES

DRAWING SUPPORT	Drawing clean-up / redlines / as-builts / fabrication and shop drawings
MECHANICAL CAD	Parts / assemblies / exploded views / packaging studies / design development
FABRICATION	Sheet-metal details / cut profiles / CNC-ready DXF / material and process notes
FILE HANDOVER	DWG / DXF / STEP / STL / PDF / agreed native formats

07 / START A PROJECT

Need drawings that are clear, controlled and ready for the next step?

I support contractors, fabricators, product teams and individual clients with practical CAD drafting and mechanical design work - from a single drawing clean-up to a coordinated package of parts and assemblies.

SEND WITH YOUR ENQUIRY

REFERENCE	Sketches, photographs, marked-up drawings, existing CAD or a sample product.
REQUIREMENTS	Key dimensions, material, manufacturing method, applicable standard and intended use.
OUTPUT	Required native format and exports - for example DWG, DXF, STEP, STL or PDF.
TIMING	Target review date, final deadline and any staged issue dates.
CONFIDENTIALITY	Confirm whether an NDA, anonymisation or restricted file-sharing method is needed.

CONTACT

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AVAILABILITY Available for selected freelance drafting and mechanical design projects. Confidential work can be handled under NDA.